Rebuilding the Batter's Brain:

Evaluating Swing Decisions through Kinematic LSTM Perception and Hierarchical Bayesian Inference

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Ground Principles

The Geometry of the Strike Zone

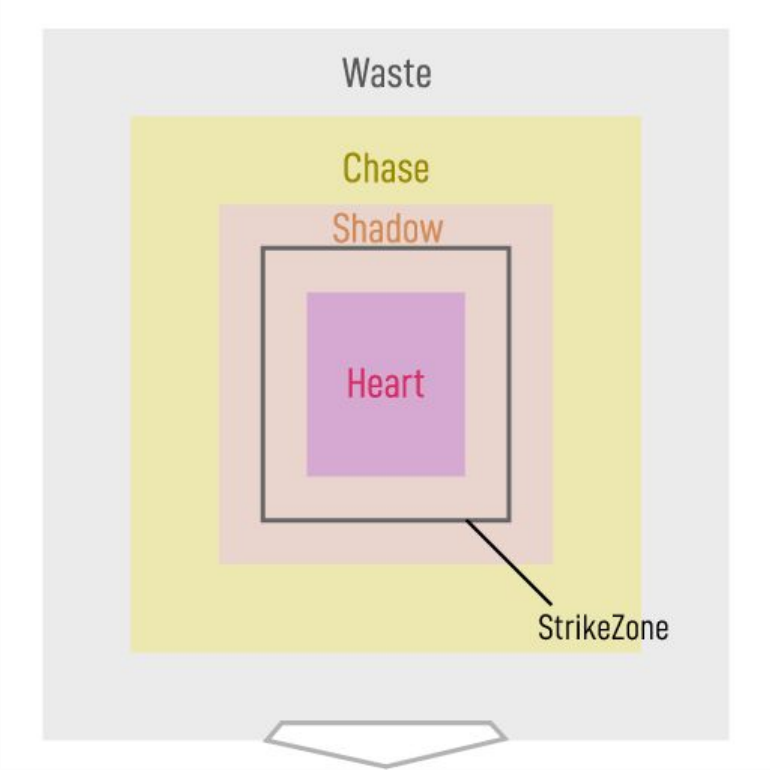


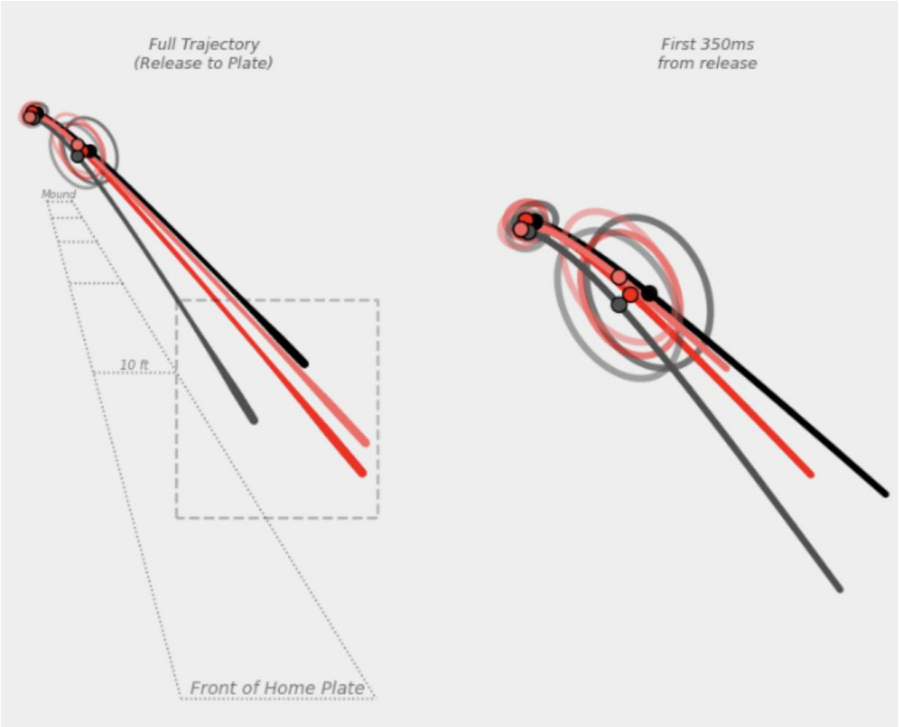
Plate Zone	No. of Pitches	% of Pitches	Swing Rate	Miss Rate
Heart of Plate	201288	25.4%	70.6%	12.6%
Shadow Zone	338806	42.8%	51.7%	23.4%
Chase Zone	182078	23.0%	23.6%	51.9%
Waste Zone	70352	8.9%	6.6%	86.7%

	Batted Ball Events Only		
Plate Zone	xBA	xwOBA	HR Rate
Heart of Plate	0.347	0.412	6.3%
Shadow Zone	0.304	0.326	2.9%
Chase Zone	0.246	0.238	0.9%
Waste Zone	0.196	0.180	0.0%

Reading Pitches

Batters predict pitch location using:

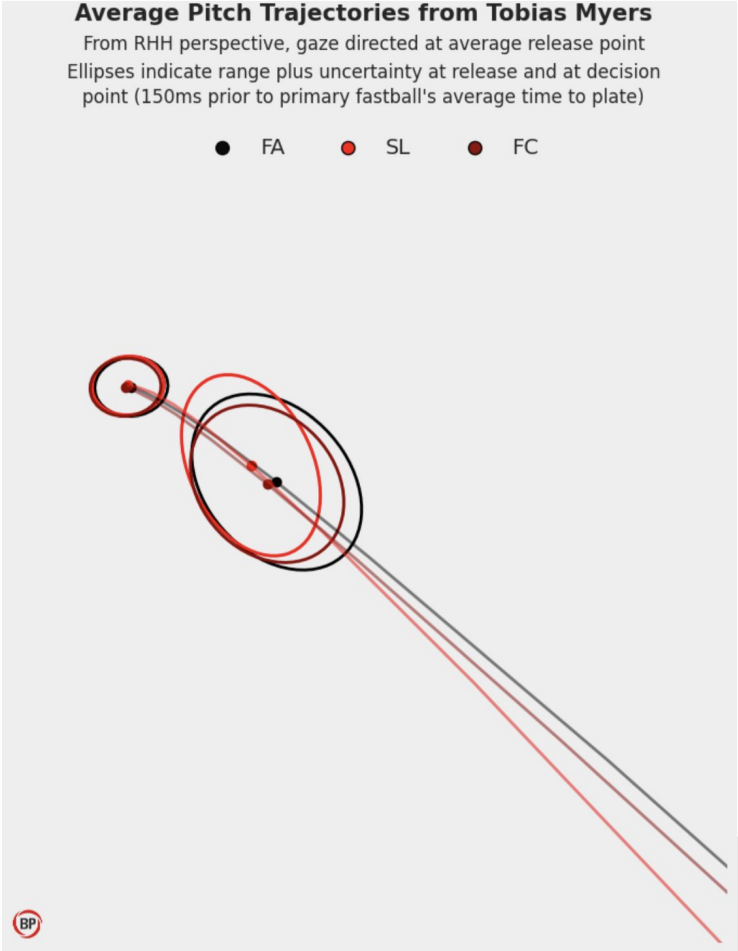
- Pitcher Handedness
- Arm Angle
- Release Angle (~50ms after release)
- Pitch Velocity (increase in size of ball)
- Trajectory Arc (deviation from straight line from release to decision point)
- Distance Traveled to Decision Point



Making Swing Decisions

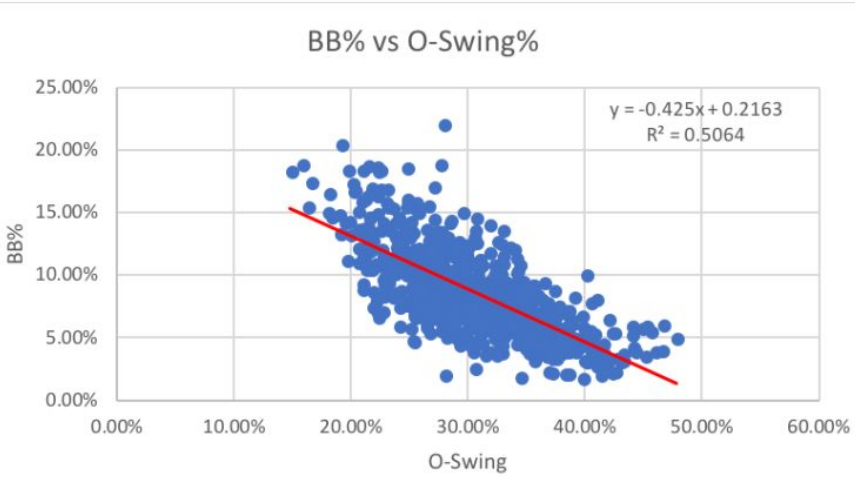
- **Total Flight Time (T_f):** ≈ 400 ms
- **Swing Execution (T_s):** 150 ms
- **Biomechanical/Neural Delay (T_b):** $50 - 75$ ms
- **Effective Decision Window:**
  $T_{decision} = 400 - (150 + 75) = \mathbf{175}$ **ms**

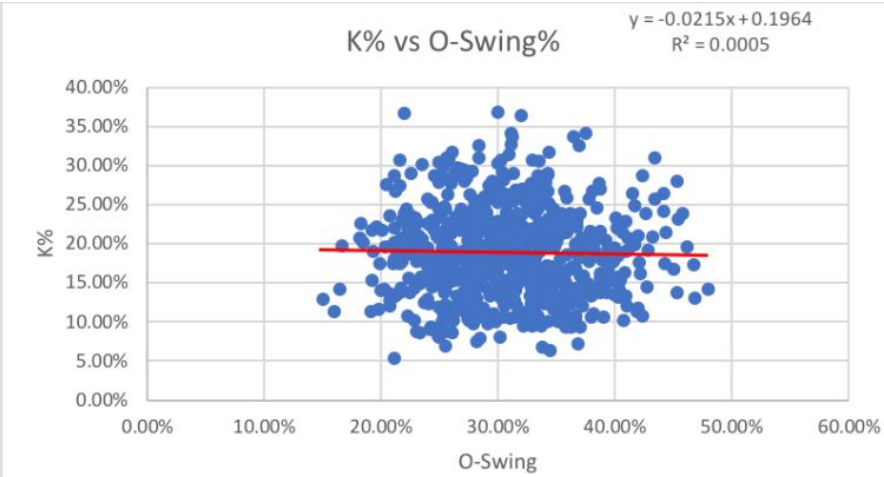
The batter hits a physical Point of No Return at ~40% of the flight distance

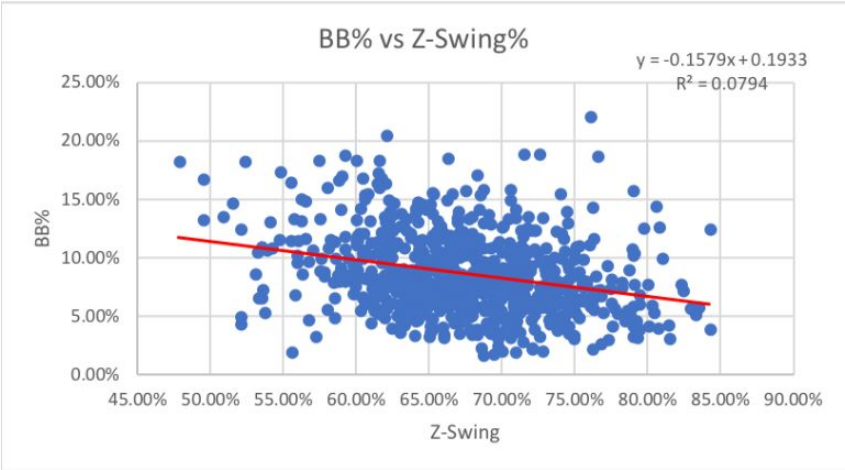


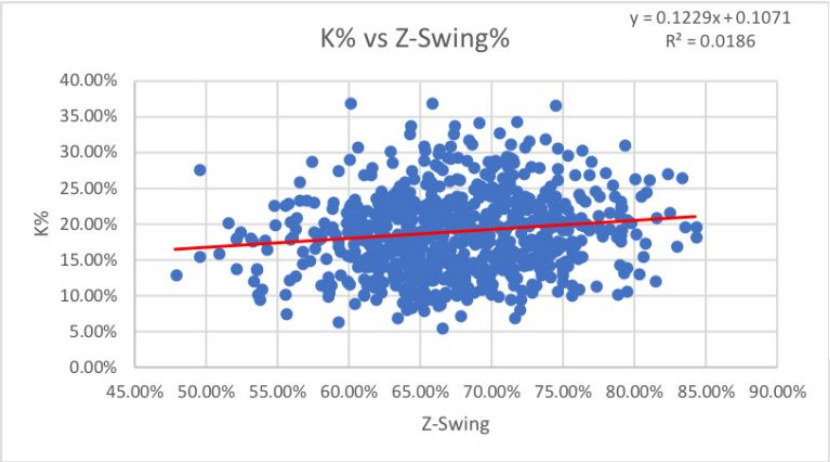
Evaluating Swing Decisions

- Zone-Swing%: % of pitches seen inside the strike zone swung at
- O-Swing%/Chase%: % of pitches seen outside the zone swung at
- Traditional Wisdom:
  - High Z-Swing% = Good
  - High Chase% = Bad


BB% vs O-Swing%
y = -0.425x + 0.2163
R² = 0.5064
BB%
O-Swing
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
0.00%
10.00%
20.00%
30.00%
40.00%
50.00%
60.00%


K% vs O-Swing%
y = -0.0215x + 0.1964
R² = 0.0005
K%
O-Swing
40.00%
35.00%
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
0.00%
10.00%
20.00%
30.00%
40.00%
50.00%
60.00%


BB% vs Z-Swing%
y = -0.1579x + 0.1933
R² = 0.0794
BB%
Z-Swing
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
45.00%
50.00%
55.00%
60.00%
65.00%
70.00%
75.00%
80.00%
85.00%
90.00%


K% vs Z-Swing%
y = 0.1229x + 0.1071
R² = 0.0186
K%
Z-Swing
40.00%
35.00%
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
45.00%
50.00%
55.00%
60.00%
65.00%
70.00%
75.00%
80.00%
85.00%
90.00%

01

Kwan vs. Stowers

A Case Study

Stowers vs. Kwan

2025 | MIA



Kyle Stowers

2025 | CLE



Steven Kwan

Overall Stats		
4.0	**fWAR**	3.2
457	**PA**	**693**
.288	**AVG**	.272
.368	**OBP**	.330
.544	**SLG**	.374
.912	**OPS**	.705
115	**H**	**170**
25	**HR**	11
73	**RBI**	56
5	**SB**	**21**
61	**R**	**81**
27.4%	**K%**	**8.7%**
10.5%	**BB%**	7.9%
1	**DRS**	**22**

By: Sox_Savant

Data: FanGraphs, Bref

2025 | MIA



Kyle Stowers

2025 | CLE



Steven Kwan

Overall Stats		
29.0%	**Chase%**	**22.8%**
30.1%	**Whiff%**	**7.9%**
27.4%	**K%**	**8.7%**
10.5%	**BB%**	7.9%
53.9%	**Swing%**	41.7%

By: Sox_Savant

Data: FanGraphs, Bref

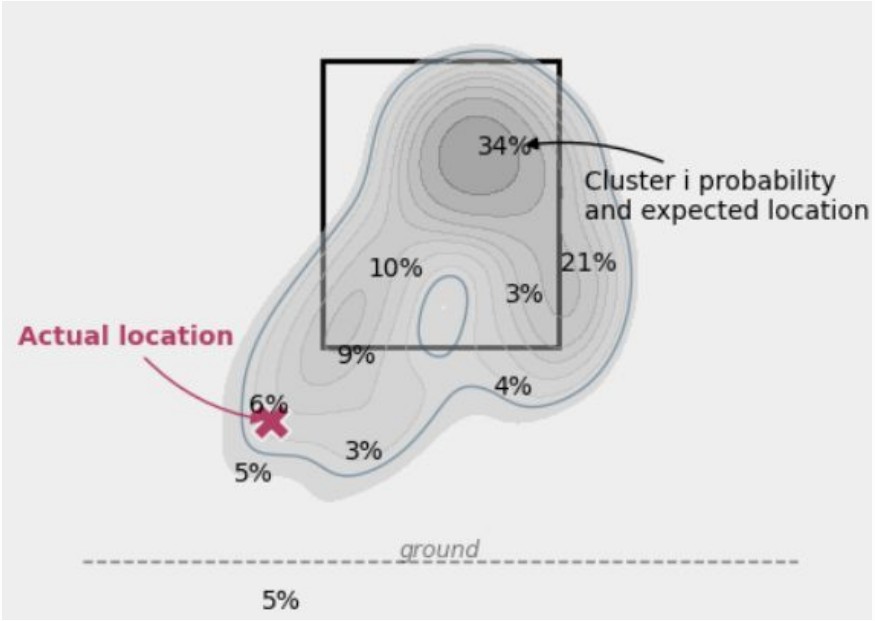
02

Breaking the Box

Why Binary Metrics Fail

Result-Oriented Logic

- Post-hoc: judges decision based on final outcome instead of information at decision point
- Ignores the process of decision & only penalizes result



The Binary Trap

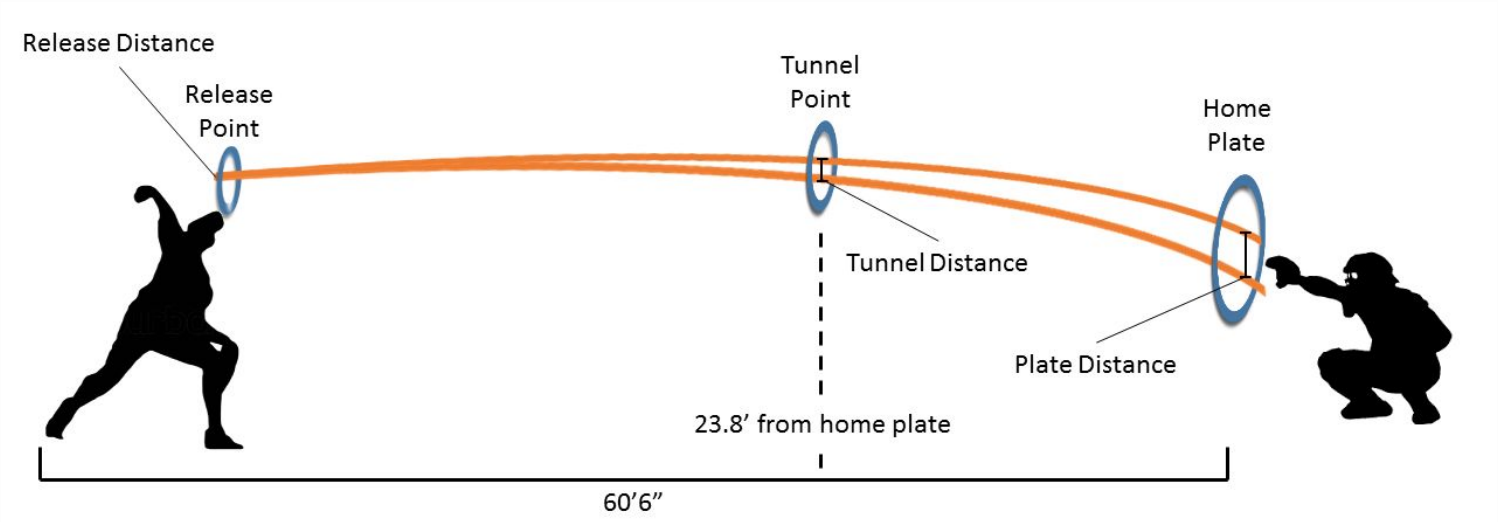
Standard discipline metrics treat the zone as a rigid box w/ 0 margin of error

- Human perception is probabilistic: batters see a “cloud” of where the ball might be, not a box
- Binary metrics don’t account for the value of “shadow” decisions which most value is derived from

Information Decay & Kinematic Uncertainty

Traditional stats assume all pitches display provide the same information

- Tunneling & late break creates different levels of perceptual difficulty
- Chase% treats a chase on a hanging slider & tunneled pitch the same



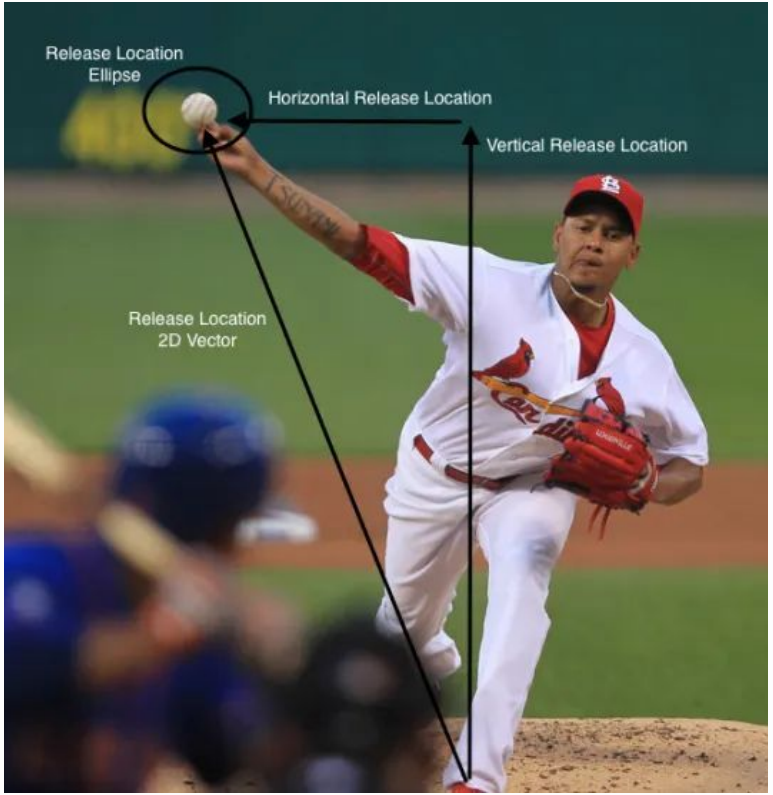
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Methodology

LSTM & Bayesian Integration

The Kinematic Inputs

Release Position (x0, z0): exact horizontal & vertical coordinates of the ball on release
Initial Velocity (vx0, vy0, vz0): the raw speed of pitch separated across all three 3D axes on release
Acceleration Vectors (ax, ay, az): constant forces acting on the ball during flight, capturing the pull of gravity, air drag, and the aerodynamic Magnus effect generated by the ball's spin



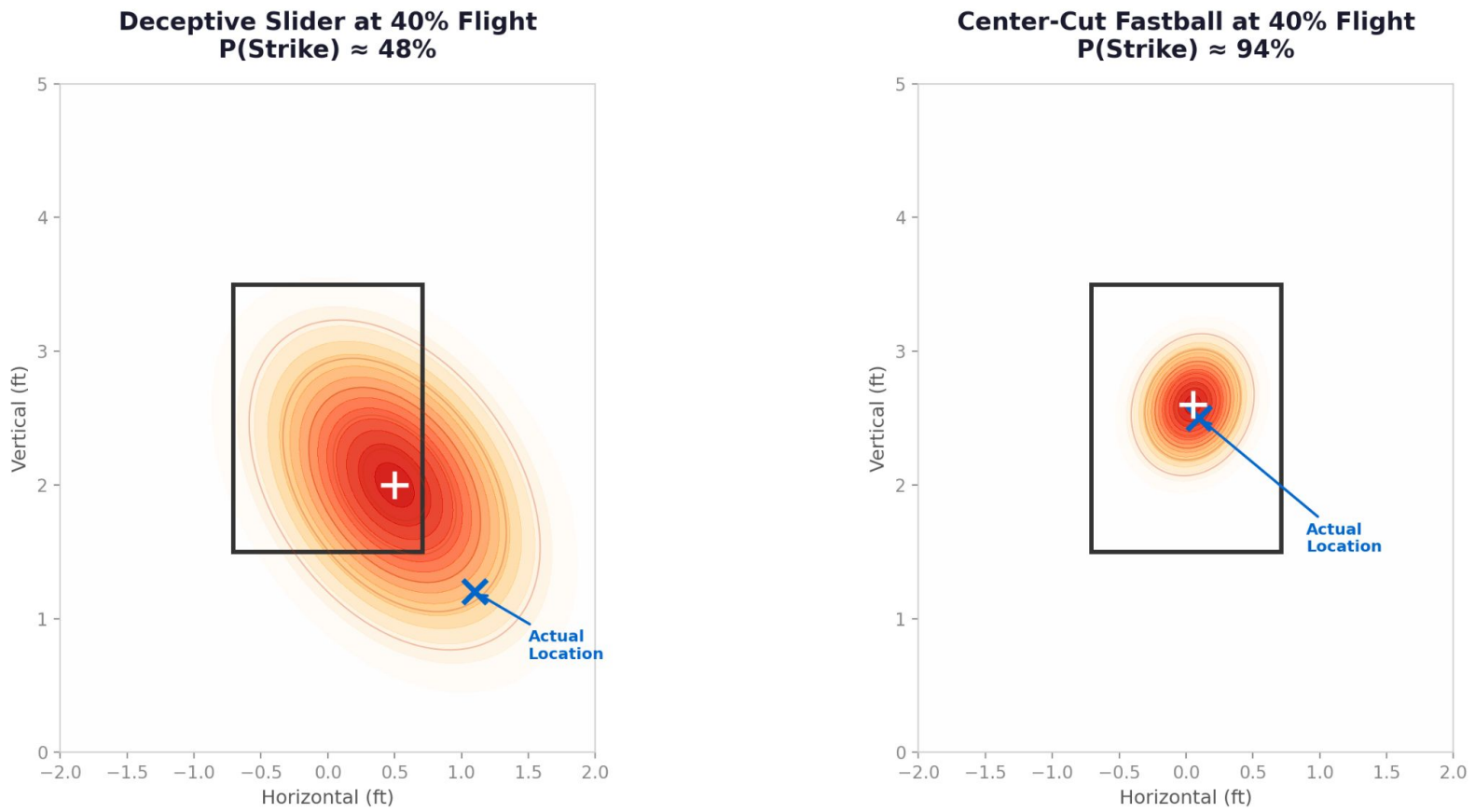
Long Short-Term Memory

- **Sequential Processing:** Designed for data that arrives one step at a time (like a pitch in flight).
- **Hidden Memory State:** Maintains an internal “memory” that updates with each new observation to learn patterns across time.
- **The Biological Analogy:** A batter’s eye functions as a *biological LSTM*—constantly updating an internal belief based on new visual fragments while carrying forward previous frames.

The Probability Cloud

- The LSTM doesn't predict a single “dot” where the ball lands. It generates a Bivariate Gaussian Distribution—a visual “cloud” of potential outcomes.
- 5 Core Parameters:
  - μ_x, μ_z (The Center): The model’s best guess for the final location
  - σ_x, σ_z (The Spread): Level of uncertainty: wider cloud = unsure
  - ρ (The Correlation): how vertical & horizontal errors interact
  - $P(strike)$: Integrating the cloud over the strike zone to estimate the probability is a strike at the 175ms decision point

The LSTM Probability Cloud at the Biological Decision Point



Grading the Decisions

Decision Value Added (DVA)

- XGBoost Model trained on 724,000+ MLB pitches to establish "expected league behavior"
- Inputs: P(Strike), count, pitcher/hitter matchup, gamestate
- Output: Positive DVA succeeds where the average hitter fails
  - Laying off a tunneled "strike-to-ball" slider

$$\text{Actual Quality} = (\text{Swing} \times P_{\text{Strike}}) + (\text{Take} \times (1 - P_{\text{Strike}}))$$

$$\text{League Quality} = (P_{\text{Swing_League}} \times P_{\text{Strike}}) + (P_{\text{Take_League}} \times (1 - P_{\text{Strike}}))$$

$$\text{DVA} = \text{Actual Quality} - \text{League Quality}$$

Hierarchical Bayesian Smoothing

- **Small Sample Noise:** batters may only see a limited amount of sliders in the shadow all season, making raw DVA volatile
- **Beta-Binomial Shrinkage:** updating a league-average tendency prior for a specific context with the batter's actual decisions
  - Higher N = growing weight of evidence, reflecting batter's unique behavior

2025 DVA Leaders — Top 10

Rank	Player	Rating	Tendency DVA	Actual DVA	Pitches
1	**Kyle Tucker**	**127.4**	+0.0627	+0.0951	2439
2	**Corey Seager**	**124.0**	+0.0512	+0.0879	1668
3	**Jahmai Jones**	**122.2**	+0.0396	+0.0892	613
4	**Matt Thaiss**	**119.5**	+0.0421	+0.0721	768
5	**Gleyber Torres**	**119.1**	+0.0450	+0.0666	2814
6	**Miguel Vargas**	**119.0**	+0.0478	+0.0632	2323
7	**Jack Suwinski**	**118.8**	+0.0371	+0.0731	806
8	**Isaac Collins**	**118.5**	+0.0449	+0.0633	1828
9	**Marcus Semien**	**118.1**	+0.0438	+0.0622	2075
10	**Ronald Acuña Jr.**	**117.8**	+0.0414	+0.0634	1741

2025 MLB Season - Min. 500 Pitches

2025 DVA Leaders — Bottom 10

Rank	Player	Rating	Tendency DVA	Actual DVA	Pitches
412	**Michael Harris II**	**76.9**	-0.0556	-0.0665	2173
413	**Jose Altuve**	**74.0**	-0.0633	-0.0748	2314
414	**Edmundo Sosa**	**73.7**	-0.0550	-0.0849	950
415	**Cody Freeman**	**73.0**	-0.0588	-0.0845	505
416	**Martín Maldonado**	**72.7**	-0.0551	-0.0898	569
417	**Jose Iglesias**	**69.5**	-0.0743	-0.0888	1378
418	**Jhonkensy Noel**	**69.1**	-0.0582	-0.1069	576
419	**Yohel Pozo**	**68.2**	-0.0679	-0.1021	591
420	**Lenyn Sosa**	**65.7**	-0.0844	-0.0996	2056
421	**Javier Báez**	**65.0**	-0.0825	-0.1053	1644

2025 MLB Season - Min. 500 Pitches

Stowers vs. Kwan — The Passivity Penalty

Kyle Stowers		Steven Kwan
114.8	**DVA Rating**	**97.1**
+0.0347	**Tendency DVA**	**-0.0105**
+0.0536	**Actual DVA**	**+0.0006**
1,712	Pitches Seen	2,829
54.0%	Overall Swing%	42.0%
29.0%	Chase% (Savant)	22.8%
77.5%	**Z-Swing% (Savant)**	**56.9%**
63.9%	Edge Swing Rate	46.8%
35.9%	**Elite Decisions**	**31.8%**
24.4%	**Terrible Decisions**	**28.0%**
+0.0296	**Avg DVA / Pitch**	**-0.0074**

2025 | MIA



Kyle Stowers

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Steven Kwan

Overall Stats		
29.0%	**Chase%**	**22.8%**
30.1%	**Whiff%**	**7.9%**
27.4%	**K%**	**8.7%**
10.5%	**BB%**	7.9%
77.5%	**Z-Swing%**	56.9%

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